Once Google, Facebook and other algorithms become all-knowing oracles, they may well evolve into agents and finally into sovereigns.[33] To understand this trajectory, consider the case of Waze – a GPS-based navigational application which many drivers use nowadays. Waze isn't just a map. Its millions of users constantly update it about traffic jams, car accidents and police cars. Hence Waze knows to divert you away from heavy traffic, and bring you to your destination through the quickest possible route. When you reach a junction and your gut instinct tells you to turn right, but Waze instructs you to turn left, users sooner or later learn that they had better listen to Waze rather than to their feelings.[34]

At first sight it seems that the Waze algorithm serves us only as an oracle. We ask a question, the oracle replies, but it is up to us to make a decision. If the oracle wins our trust, however, the next logical step is to turn it into an agent. We give the algorithm only a final aim, and it acts to realise that aim without our supervision. In the case of Waze, this may happen when we connect Waze to a self-driving car, and tell Waze 'take the fastest route home' or 'take the most scenic route' or 'take the route which will result in the minimum amount of pollution'. We call the shots, but leave it to Waze to execute our commands.

Finally, Waze might become sovereign. Having so much power in its hands, and knowing far more than we know, it may start manipulating us, shaping our desires and making our decisions for us. For example, suppose because Waze is so good, everybody starts using it. And suppose there is a traffic jam on route no. 1, while the alternative route no. 2 is relatively open. If Waze simply lets everybody know that, then all drivers will rush to route no. 2, and it too will be clogged. When everybody uses the same oracle, and everybody believes the oracle, the oracle turns into a sovereign. So Waze must think for us. Maybe it will inform only half the drivers that route no. 2 is open, while keeping this information secret from the other half. Thereby pressure will ease on route no. 1 without blocking route no. 2.

Microsoft is developing a far more sophisticated system called Cortana, named after an AI character in their popular *Halo* videogame series. Cortana is an AI personal assistant which